

Individual room air-conditioning control in high-insulation residential building during winter: A deep reinforcement learning-based control model for reducing energy consumption

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ARTICLE INFO

Keywords:

Deep reinforcement learning
DQN
Room air-conditioner
Residential building
Overheating

ABSTRACT

In recent years, the thermal insulation performance of residential buildings has been enhanced to reduce energy consumption. However, this enhancement often leads to air conditioning systems operating under ultra-low load conditions for extended periods, especially in individual rooms, which frequently results in sustained low efficiency. Additionally, during the winter, rooms tend to overheat due to the influence of solar radiation. In this study, we developed a deep reinforcement learning-based real-time & prediction full-stack control model, which can automate user-end air-conditioning control through home energy management system (HEMS). In this model, weather forecasts are utilised to mitigate overheating caused by solar radiation, thereby reducing energy consumption. Additionally, it can enhance the COP of air conditioners in low-load domains. Our successful empirical results indicate that the implementation of this model can reduce energy consumption by approximately 40% in winter.

1. Introduction

1.1. Background

With the growing global focus on climate change, achieving carbon neutrality by 2050 has become a key target for the Japanese government [1]. Energy consumption in buildings, which accounts for 20% to 40% in developed nations [2], is under increasing pressure to be reduced. This is especially true in residential buildings, where air-conditioning systems are significant energy consumers that impact household budgets. Reducing energy use in these systems is not only crucial for alleviating financial strain but also essential for achieving carbon neutrality goals.

In recent years, enhancing building thermal performance has been prioritized in research and practice to address energy challenges. Advances in building materials, design, and construction have notably improved insulation and heat resistance. In 2022, Japan's Ministry of Land, Infrastructure, Transport and Tourism introduced superior insulation levels 6 and 7 to the existing standards, aiming to reduce the thermal load [3].

However, relying solely on improving thermal insulation performance to reduce thermal loads has limited effectiveness in energy sav-

ings. This is especially evident in high-insulation residential buildings, where air-conditioning systems are forced to operate at ultra-low loads owing to the minimal heating load in winter, particularly in individual rooms where the heating load in winter is minimal, resulting in poor efficiency according to the coefficient of performance (COP) curve [4]. Moreover, during daylight hours, solar radiation can frequently lead to room overheating, resulting in unnecessary energy consumption. Therefore, to address the inefficiencies of air conditioners and enhance energy savings in high-performance residential buildings, optimizing the control of air conditioning systems is of paramount importance.

Rule-based control (RBC) is a traditional control method, wherein system operations are adjusted based on predefined rules and strategies. The advantages of RBC include its simplicity, ease of implementation, and suitability for basic energy optimisation tasks. However, its performance is constrained by the effectiveness and applicability of fixed rules, and it cannot adapt to the dynamic changes in the system. Consequently, RBC has proven to be a suboptimal approach [5]. Model predictive control (MPC) is an advanced control method that predicts the future behaviour of the system based on mathematical models, and accordingly implements control decisions. It is capable of adapting to changes in the system and its dynamic nature. Empirical results obtained

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<https://doi.org/10.1016/j.enbuild.2024.114799>

Received 9 April 2024; Received in revised form 22 August 2024; Accepted 9 September 2024

Available online 12 September 2024

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Table 1
DRL control in residential buildings.

Ref.	Target system	Control action	Algorithm	Environment	Weather forecast
[16]	AC	Temperature set point	DQN	EnergyPlus	×
[17]	AC, Window	ON/OFF	DQN	Grey-box model	×
[18]	AC, Air-cleaner, Window	ON/OFF	DQN	Grey-box model	×
[19]	AC	Temperature set point	DQN	Grey-box model	×
[20]	AC	Temperature set point	DQN	Black-box model	×

from practical applications in buildings have demonstrated its effectiveness [6–8]. However, the computational complexity of MPC makes it difficult to deploy on low-power, low-computational-capacity HEMS systems. Additionally, the efficacy of MPC controllers highly dependent on the accuracy of the system models used. Inaccuracies in these models can lead to suboptimal performance in practical applications. As elucidated by Afram et al. [9], MPC may not be apt for systems requiring the maintenance of a stable temperature, underscoring the method's limitations in adapting to real-world complexities.

In recent years, with the emergence of big data, powerful computing capabilities, and algorithmic advancements, machine learning-based control methods for building HVAC systems have been extensively explored and applied. Notably, reinforcement learning approaches, which learn control strategies from historical data or energy simulation and then directly deploy the model to the target building, have significantly enhanced the efficiency of practical deployment. [10]

1.2. Reinforcement learning-based HVAC system control model

Deep reinforcement learning (DRL) is a machine learning-based control method that maximises control rewards through interactions between the agent and environment [11]. In non-residential buildings, numerous studies have substantiated that the application of DRL for controlling HVAC systems can result in significant energy savings [12–15]. However, in the context of residential buildings, the practical application of DRL control is challenging as the load and the COP are more significantly affected by external weather conditions, and users cannot directly control the input heat of the air conditioner. Research on DRL control in residential buildings is summarized in Table 1.

Huchuk et al. [16] conducted a year-long simulation on 500 residential models in EnergyPlus, discovering that, compared to traditional heuristic control methods, the model predictive controller significantly reduced the total cost of operation and discomfort. However, the RL approach underperformed due to sparse observational data and restricted exploration range. Chen et al. [17] introduced a RL control strategy to optimize decision-making for HVAC and window systems. In case studies conducted in hot and humid Miami and warm and mild Los Angeles, this approach reduced HVAC energy consumption by 13% and 23%, discomfort hours by 62% and 80%, and high humidity hours by 63% and 77%, respectively, compared to heuristic control. Similarly An et al. [18] conducted pioneering research into the reinforcement learning control of air conditioners, air cleaners, and window switches, affirming the promising application of DRL in residential buildings. However, the aforementioned study primarily focused on the rated capacity of air conditioners without considering the significant impact of their Coefficient of Performance (COP) on energy consumption. [19] utilized a DRL controller to manage air conditioning systems, aiming to enhance energy efficiency. Additionally, the study incorporated a simplified air conditioner COP model described in [21]. However, in real-world residential settings, the COP of air conditioners is influenced by external weather conditions and indoor temperatures. [20] utilized Bayesian Convolutional Neural Networks (BCNNs) to model air conditioning systems and trained RL agents in a black-box simulation environment. In this approach, energy consumption is calculated based on the black-box model. However, in practice, training a high-accuracy model requires a large amount of high-quality data, which can be time-consuming and

resource-intensive. Furthermore, none of the aforementioned studies on residential buildings have accounted for the uncertainty in long-term future weather variations.

In summary, the following critical gaps have been identified in the existing literature, which underscore the need for further in-depth investigation and research:

- The Coefficient of Performance (COP) of air conditioners is a critical factor in energy consumption. However, existing literature often lack a simulation environment that includes a detailed COP model and fail to provide sufficient explanations for their optimization strategies.
- There is a gap in the literature regarding the application of RL in high-insulation residential buildings. Particularly during winter, these buildings can experience overheating due to solar radiation, underscoring the need to consider the uncertainty of future weather conditions. However, current studies have not adequately explored the impact of this uncertainty.
- Existing research has extensively demonstrated the effectiveness of RL. However, discussions on the challenges faced during actual deployment are scarce. For instance, the platforms used for deployment, the methodologies applied, and the collection of training data are seldom addressed in the literature.

To address these gaps, we have developed a comprehensive network strategy, inspired by the research on transfer learning conducted by Xu et al. [22]. This approach integrates a detailed model of the COP of air-conditioning systems developed by the Central Research Institute of Electric Power Industry (CRIEPI), Japan [28], taking into account their impact on energy consumption. This approach enables the precise mapping of user settings to the input heat of the air-conditioning system through a Home Energy Management System (HEMS) at the user end, thereby facilitating the control based on strategy made by DRL and enhancing the interpretability of the strategies. Furthermore, we integrated the predictive data from weather forecasts into the DRL model, which enables the intelligent adjustment of the air-conditioning system not only based on real-time state but also on predictive data, aiming to mitigate overheating owing to solar radiation and enhance the efficiency of air-conditioning operation under low-load conditions. Compared with traditional DRL methods, our approach can achieve significant improvements in energy use.

The major contributions of our study are as follows:

- (1) In our research, a DRL-based control model was crafted, meticulously considering the complex effects of external weather, indoor conditions, and load rates on the Coefficient of Performance (COP) of air conditioning systems. This model effectively utilizes real-time data and uncertain weather forecast information to manage the heat input control of air conditioning systems, thereby achieving energy savings.
- (2) We developed a full-stack control model that maps planned input heat to user-end air conditioning settings through supervised learning methods. This enhances the interpretability of RL control strategies and enables intelligent management of air conditioning systems via HEMS.
- (3) We have detailed the approaches to data augmentation as well as the methods used for deployment. Subsequently, we tested the

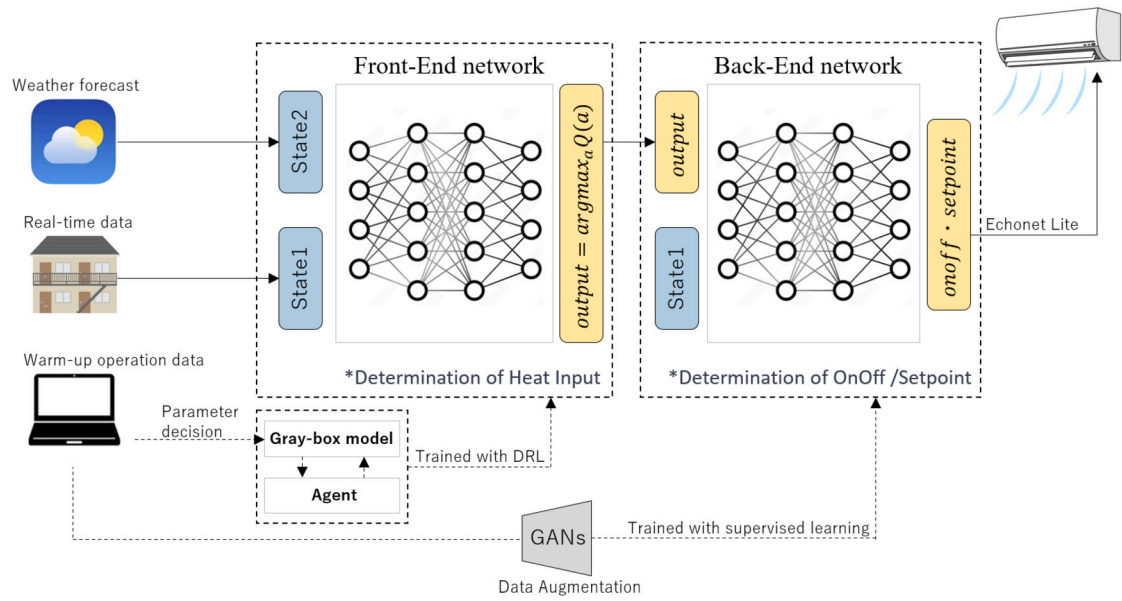


Fig. 1. Our approach.

control model in a full-scale high-insulation laboratory. Empirical results indicate that our model can achieve approximately 40% energy savings in real-world applications.

The structure of the remaining sections of this study is as follows: Section 2 describes the methodology employed in this study. Section 3 is dedicated to a case study based on simulations, where the target building is conceptualised as a rooftop full-scale laboratory. Subsequently, in section 4, we conducted real-world testing to validate the effectiveness of the offline learning methods. Lastly, section 5 serves as the conclusion and outlines potential future work in this field.

2. Methodology

Our approach is shown in Fig. 1. The control model consists of two parts: the front-end network and the back-end network. The front-end network optimizes the input heat of the air conditioning system using DRL. While the back-end network, with an emphasis on building performance and employing supervised learning, maps the planned input heat to the air conditioning settings, realizing a full-stack network control.

The specific training steps are as follows:

- (1) A simulation environment based on a grey-box model is created using warm-up operation data. The model parameters are determined by a Genetic Algorithm (GA).
- (2) Based on this simulation environment, the front-end network is trained using DRL to learn the control strategy for the input heat of the air-conditioning system.
- (3) By employing Generative Adversarial Networks (GANs) for data augmentation on warm-up operation data, the back-end network is then learned through supervised learning methods to achieve user-end control of the planned input heat for the air-conditioning system.

2.1. Front-end network

2.1.1. Establishment of the simulation environment

The front-end network is trained using reinforcement learning, typically prioritizing the offline training mode in the application of DRL due to its provision of safety and efficiency. The selection of a simulation environment plays a pivotal role throughout the training process. In thermal simulations of buildings, the simulation environments are

categorised into black-, grey-, and white-box models. Considering the difficulty in fully expressing complex building environments through white-box models, and the cost implications associated with the substantial real-world data required for training an accurate black-box model, we employed a grey-box model (RC model) [23]. The RC model is analogous to electrical resistance and capacitance to thermal resistance and capacitance, respectively. A wealth of research on the RC model has underscored its efficacy and precision in replicating the thermal conditions within indoor environments [24,25]. Consequently, we employed the RC model to accurately simulate the temperature response of the target building.

We considered the thermal resistance and thermal capacitance of the walls, as well as the thermal resistance and thermal capacitance of indoor air. We employed a 2R2C model to represent the thermal response of the target building. The model operates within the framework of linear state-space models, offering a methodical approach to understanding how buildings react to thermal inputs over time. The equations of the model are expressed as follows (Equation (1)):

$$\begin{cases} C_i \frac{dT_i}{dt} = \frac{1}{R_i}(T_e - T_i)dt + \Phi_h dt + A_i \Phi_g dt, \\ C_e \frac{dT_e}{dt} = \frac{1}{R_i}(T_i - T_e)dt + \frac{1}{R_o}(T_o - T_e)dt + A_e \Phi_g dt, \end{cases} \quad (1)$$

where T_i , T_e , and T_o denote the temperatures of the indoor environment, building envelope, and outdoor environment, respectively. The temperature T_e is associated with the thermal mass of the opaque surfaces within the envelope and does not correspond to a specific spatial coordinate within the envelope. The model encompasses two states: T_e (unobserved) and T_i (observed). The term Φ_h (W) represents the internal heating power, whereas Φ_g (W/m^2) denotes the global horizontal solar irradiance. The coefficient R_i (K/W) is the thermal resistance between the indoor air temperature and envelope, and R_e represents the resistance between the envelope and ambient air. The heat capacitances of the interior and envelope are represented by C_i and C_e (J/K), respectively. The coefficients A_i and A_e (m^2) correspond to their solar gain coefficients.

Based on the measured data, we can utilize a Genetic Algorithm (GA) to determine the optimal values of the aforementioned parameters.

2.1.2. Markov Decision Process (MDP)

In our study, we analogize the control of air conditioning input heat as a prototypical reinforcement learning issue, wherein each reinforcement learning challenge can be represented in a Markov Decision Process (MDP) [26]. The MDP encompasses three key elements: states s , actions a , and rewards r . State transition P from the MDP satisfies the following Markov assumption:

$$P(s_{t+1}|s_t, a_t, r_t, \dots, s_0, a_0, r_0) = P(s_{t+1}|s_t, a_t) \quad (2)$$

Here, s_t represents the state at time t , a_t represents the action taken at time t , and r_t represents the reward received at time t . This equation illustrates the Markov property, indicating that the future state s_{t+1} depends only on the current state s_t and action a_t , regardless of the previous history of states, actions, and rewards. In our study, states s , actions a , and rewards r are detailed as follows:

States s

To bolster the robustness of our model against the complexities and variabilities of external atmospheric conditions, our methodology encompasses the assimilation of real-time operational data, denoted as state1, with predictive forecast data, referred to as state2. Shin et al. [27] employed GRU to extract features from predictive data. However, considering the complexity of GRUs, which are not conducive to being implemented on low-power, compact computers, we directly incorporated the prediction data as state2. The specific details are encapsulated in the following equation:

$$\begin{aligned} \text{state1}_i &= [t_{\text{out},i}, t_{\text{room},i}, \text{onoff}_i, \text{solar}_i, h_i], \\ \text{state2}_i &= [t_{\text{out},h_i+1}, t_{\text{out},h_i+2}, \dots, t_{\text{out},h_i+12}, \text{solar}_{h_i+1}, \text{solar}_{h_i+2}, \dots, \\ &\quad \text{solar}_{h_i+12}] \end{aligned} \quad (3)$$

where, i denotes the time step at the i^{th} moment, t_{out} represents the outdoor temperature, t_{room} indicates the room temperature. The variable onoff reflects the current operational status of the air conditioner, with 0 for off and 1 for on. The term solar stands for the global horizontal solar irradiation, and h signifies the current absolute time.

Action a

The control action, defined as the input heat of the air-conditioning system (L), is discretised in intervals of 100 W, enabling a granular approach to thermal management within the predefined operational range. The specific details are encapsulated in the following equation:

$$a = [0, L_{\min}, L_{\min} + 100, L_{\min} + 200, \dots] \quad (4)$$

Where $L_{\min}$ denotes the minimum capacity of the air-conditioning system.

Reward r

In high residential buildings, the heating load during winter is minimal, which leads to the operation of air conditioning systems in the low COP area. This is primarily because the compressor of the air-conditioning system tends to frequently cycle on and off. Furthermore, the influence of solar radiation makes the rooms prone to overheating. To mitigate overheating and reduce energy consumption, as well as enhance the COP of the air-conditioning system, we designed a reward function. This function incorporates rewards induced by comfort (R_1), penalties attributed to energy consumption are calculated through the air-conditioning COP under various load rate (R_2), and penalties associated with overheating occurrences (R_3). We utilised a standard air-conditioning model, developed by the Central Research Institute of Electric Power Industry (CRIEPI), Japan, to accurately calculate the electricity consumption based on the COP [28]. The model utilizes hyperparameters such as rated capacity, minimum and maximum capabilities, and their corresponding power consumptions (provided by the manufacturer) to estimate the ratio of the actual COP to the theoretical COP

of the air conditioner at different load rates. Based on this, the model also considers the air flow rate and the coil's bypass factor, as well as the indoor and outdoor conditions, to estimate the temperatures of the evaporator and condenser, thereby calculating the near-actual COP of the air conditioner under various conditions.

The reward comprises the aforementioned components R_1 , R_2 , and R_3 , adjusted through specific weights w_1 , w_2 , and w_3 . The exact equation is as follows:

$$\text{Reward} = w_1 \times R_1 + w_2 \times R_2 + w_3 \times R_3 \quad (5)$$

2.1.3. Deep Q-network (DQN)

To address the reinforcement learning challenge, we specifically employed the deep Q-network (DQN) method [29], based on Q-learning. DQN is adept at discerning optimal actions from within the complex, high-dimensional spaces characteristic of air-conditioning control systems. Agents continually refine their understanding of action values, or Q-values, through persistent exploration within the environment. The Q-value update is governed by the following equation:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a' \in A} Q(s', a') - Q(s, a) \right] \quad (6)$$

where α is the learning rate, γ is the discount factor, r is the reward, s and s' represent the current and next state, respectively, and a and a' denote the current and subsequent actions.

By leveraging a neural network to approximate the Q-function, the system can effectively forecast the impact of various actions, enabling the realization of strategies to conserve energy.

A Python-based agent was developed to facilitate the interaction with this environment and iteratively update the Q-values. The specific process is shown in Fig. 2.

2.2. Back-end network

As users cannot directly control the heat input of air-conditioning systems, we employed a back-end network to learn the relationship between the set temperature and heat input. The features include the real-time operational data (state1) and the planned heat input of the air-conditioning system, wherein the set temperature served as the label. Notably, this constitutes a supervised learning classification task, necessitating the learning of the back-end network through empirical data. However, in newly constructed residences, warm-up operations are required to collect data. Qian et al. [30] found that when using an artificial neural network (ANN) for short- and medium-term cooling load forecasting, the prediction accuracy of using short-term weekly data as training and validation data for full cold-season forecasting was significantly lower than that of using full-year data to build the model. However, prolonged warm-up operations could be detrimental to the cost-effectiveness. Owing to the effectiveness of GANs in data augmentation [31–33], we utilised them to augment the limited warm-up operation data, enhancing the data utilisation efficiency and reducing warm-up operation costs.

GANs are a novel approach to generative modelling, comprising two neural networks contesting with each other in a game ([34]). The objective function of GANs is expressed as follows:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (7)$$

where x represents real data sampled from the data distribution $p_{\text{data}}(x)$, z is a noise vector sampled from a predefined noise distribution $p_z(z)$, $G(z)$ denotes the data generated by the generator network from noise z , $D(x)$ is used to estimate the probability that real data x are authentic, as assessed by the discriminator network, and $D(G(z))$ is used to estimate the probability that generated data $G(z)$ are real, according to the discriminator. The generator is trained to maximise the likelihood of the discriminator making a mistake, equivalent to minimising

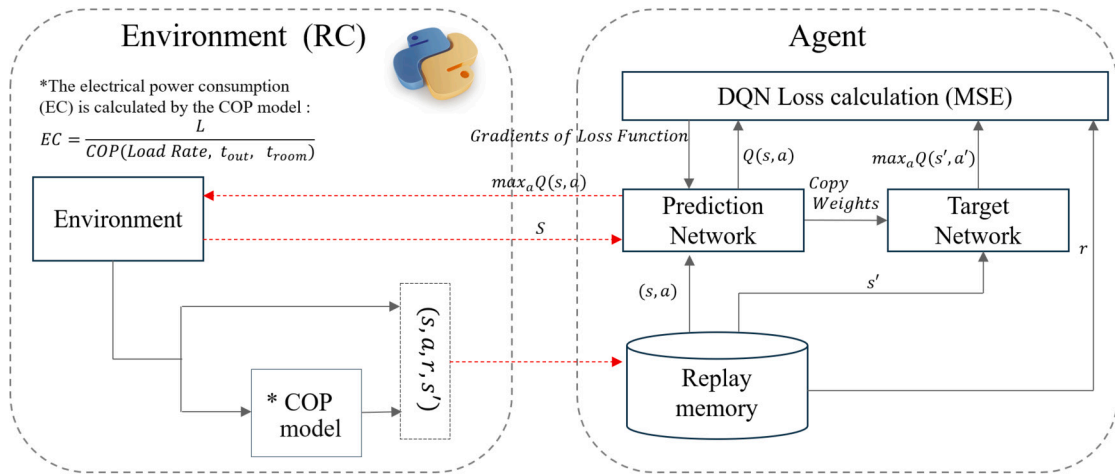


Fig. 2. Training framework of the DQN.

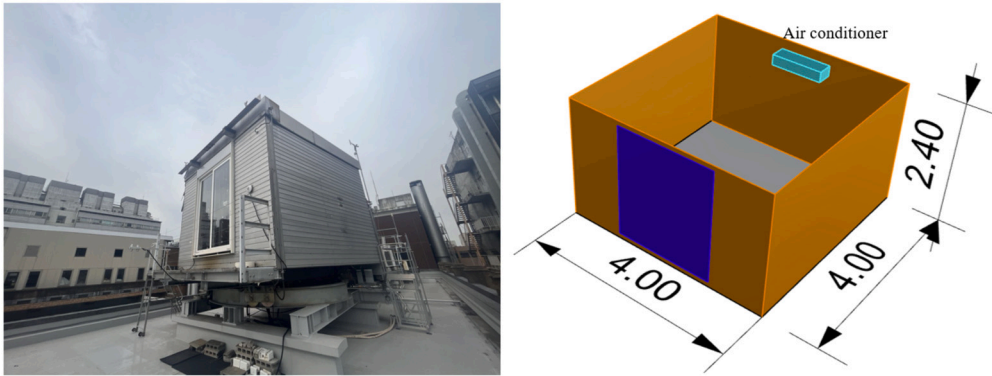


Fig. 3. View of the target building.

$\log(1 - D(G(z)))$, whereas the discriminator strives to accurately distinguish between genuine and generated data, thus maximising $V(D, G)$. The training of GANs involves iterative updates to the discriminator and generator in a minimax game, converging to an equilibrium wherein the data produced by the generator are indistinguishable from real data, rendering the discriminator unable to effectively differentiate between the two.

Considering the use of GANs for data augmentation in this study, we leveraged their capability to produce high-quality synthetic data. This method can effectively overcome the constraints associated with the need for extensive preliminary operations to collect sufficient training data. By enriching the dataset with synthetically generated samples that reflect the complex dynamics of air-conditioning systems, we enhanced the model's generalisation and performance.

3. Case study

3.1. Overview of the target building

The target building was the outdoor full-scale laboratory located on the rooftop of Engineering Building No. 1 at the University of Tokyo (Fig. 3). The insulation performance of the laboratory and experimental conditions are presented in Table 2. The orientation of the windows is southward, and there are no shading devices for solar radiation. The room's air exchange rate was set at once per hour.

In this experiment, we defined the overheating condition as a room temperature in excess of 28 °C. When the room temperature surpasses 28 °C, the fan coil unit (FCU) is automatically activated to maintain the temperature at 28 °C, simulating actions, such as the opening of windows and natural ventilation for cooling. Additionally, by measuring the

Table 2

Summary of target building.

Item	Value
Floor area S_{floor}	16.00 [m ²]
Floor height H	2.40 [m]
Volume V	38.4 [m ³]
U-Values U_a	0.3 [W/(m ² ·K)]
Window area S_{window}	3.22 [m ²]
Solar heat gain rate η_{lw}	0.83

cooling heat of the FCU, we can obtain the actual value of the room's overheating quantity.

To maintain the room temperature at the set point during winter, we employed a room air conditioner with a rated heating capacity of 2.5 kW, and a minimal heating capacity of 0.5 kW.

3.2. Resistance-capacitance model for the target building

To determine the parameters of the 2R2C model (Fig. 4), a five-day calibration campaign, with a measurement interval of 1 min, was conducted in the target building in January. We collected external air data and solar radiation data. Additionally, we gathered data on the air conditioner's inlet and outlet temperatures, as well as airflow data, to calculate the heat input. To prevent system instability and inefficiency caused by either excessively short or long control intervals, the 2R2C model was designed with a prediction interval of 20 min, necessitating the averaging of the original data over this period. The data from the first four days served as the training set, and the parameters of the 2R2C model were optimised by minimising the mean squared error (MSE)

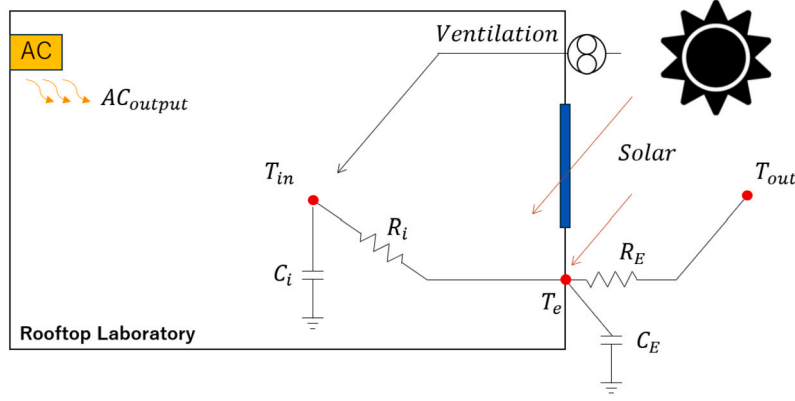


Fig. 4. The 2R2C model.

Table 3
Parameters of the 2R2C model.

Parameter Description	Value
R_i (K/W)	0.0055797
C_i (J/K)	14527
R_e (K/W)	0.15711
C_e (J/K)	25829

(Equation (8)) between the measured values y_i and predictions $\hat{y}_i$ of the 2R2C model by utilising a genetic algorithm (GA).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (8)$$

The parameters optimized through GA are presented in Table 3. Subsequently, the accuracy of the calibrated 2R2C model was validated using the data from the fifth day. Comparative analysis revealed that the model showed an MSE of 0.03, thereby demonstrating its capability for high-precision temperature reproduction.

3.3. Front-end network training

3.3.1. Design of reward function

In Chapter 2, we introduced the reward function as being composed of R_1 , R_2 , and R_3 . In this section, we specifically discuss the design methodology of R_1 , R_2 , and R_3 .

First, we defined the temperature range of 22 to 26 °C as the comfort zone. In cases with significant solar radiation, overheating can be mitigated by pre-emptively switching off the air conditioning in the morning, allowing the room temperature to slightly decrease. We hypothesise that inhabitants wake up at 9 A.M. If the room temperature falls below the lower limit of the comfort zone, namely, 22 °C, upon waking, it may cause discomfort owing to cold. Therefore, our strategy aims to reduce the room temperature prior to the awakening of residents, targeting a temperature level slightly below 22 °C to both alleviate overheating and decrease energy usage. The reward for comfort (R_1) is calculated using the following equation:

$$R_1 = \begin{cases} 500 & \text{if } 22 < T_{\text{room}} < 26 \\ 300 & \text{if } 20 < T_{\text{room}} < 22 \text{ and } 5 \leq h < 9 \text{ and } t_0 = \text{OFF} \\ -500 & \text{if } 20 < T_{\text{room}} < 22 \text{ and } 5 \leq h < 9 \text{ and } t_0 = \text{ON} \\ -200 & \text{otherwise} \end{cases} \quad (9)$$

We posit that maintaining the room temperature within the designated comfort zone yields a reward of 500. To discourage overheating, we endorse the early deactivation of the air-conditioning system, which, if it results in a room temperature marginally below 22 °C, secures a reward of 300. Conversely, to avoid the indiscriminate shutdown of the

air-conditioning system merely for energy savings, a penalty of -500 is imposed if the air conditioner is reactivated when the room temperature is below 22 °C. For any scenario where the room temperature deviates from the comfort zone, a penalty of -200 is enforced. These values were calibrated based on the relative scale of electricity consumption through trial and error.

Subsequently, we define the penalty term R_2 based on the electrical power consumption (P). Owing to the influence of the COP on the energy consumption of the air conditioner, the electrical power required is derived from the COP model [28], which takes into account the characteristic values of the air conditioner, the room temperature, the ambient temperature, and the input heat of the air conditioner.

If the air conditioner is operating at both the current time step t_1 and previous time step t_0 , R_2 is equal to the power consumption. However, if the air conditioning is turned on at the current time step t_1 , but was switched off at the previous time step t_0 , owing to the high energy consumption during startup, we define R_2 as twice the power consumption. The specific expression for R_2 is as follows:

$$R_2 = \begin{cases} -2 \times P & \text{if } t_1 = \text{ON and } t_0 = \text{OFF} \\ -P & \text{otherwise} \end{cases} \quad (10)$$

Finally, we define the reward function R_3 focused on alleviating overheating. When overheating occurs, a penalty is incurred, expressed as follows:

$$R_3 = -1000 \quad (11)$$

From the above, the reward is calculated by (Equation (5)). We established the weight coefficients w_1 and w_3 as 0.5, and w_2 as 1 through a process of tuning and experimentation.

3.3.2. Exploring the learning process and outcomes of DQN

To achieve the early shutdown of the air-conditioning system in response to solar radiation, thereby mitigating overheating, the need for such pre-emptive actions must accurately be determined. This determination involves the prediction of future conditions. Traditional DQN models primarily rely on the current environmental state for decision-making. However, this approach may not always be effective in predicting future environmental changes, especially regarding weather variations. Basing future state predictions solely on the current state can lead to an inadequate predictive capability in the state transition matrix. To overcome this limitation, we employed weather forecasts as a key component of our DQN model. Consequently, the state transition matrix no longer depends solely on the current environmental state but can incorporate both internal and external information, leading to a more accurate reflection of future state changes. This enhancement enables our model to more effectively predict future states, facilitating more rational decision-making based on these forecasts.

Table 4
Hyper-parameters of the DQN model.

Parameter	Description
<i>policy</i>	Neural network policy type ('MlpPolicy')
<i>learning rate</i>	Learning rate (1×10^{-3})
<i>buffer size</i>	Size of the replay buffer (2000000)
<i>batch size</i>	Batch size (512)
<i>gamma</i>	Discount factor (0.9)
<i>train freq</i>	Training frequency (256 steps)
<i>gradient steps</i>	Number of gradient steps (256)
<i>exploration initial eps</i>	Initial exploration probability (0.3)
<i>exploration final eps</i>	Final exploration probability (0.05)



Fig. 5. Reward convergence over training steps. (For interpretation of the colours in the figure(s), the reader is referred to the web version of this article.)

In our study, we developed three scenarios for comparative analysis: RL1, RL2, and RL3. RL1 was trained using the rated capacity of the air-conditioning system, while RL2 employed a detailed COP energy consumption model for training. RL3 also utilized the detailed COP model but further incorporated future weather conditions into the analysis. Both RL1 and RL2 were trained using only State1, which included the current indoor temperature, outdoor temperature, global solar radiation, the on/off status of the air-conditioning system, and the current time (hour). In contrast, RL3 included both State1 and State2, with State2 encompassing predicted outdoor temperature and solar radiation information for the next 12 hours. Notably, these data were compressed into a single scalar value within the range of 0 to 1 using the sigmoid function. Finally, the results of these three scenarios were compared against two baseline scenarios where the set temperature remained constant (22 and 23 °C).

By establishing these three scenarios, we aimed to evaluate the impact of employing a detailed COP model of the air conditioner and integrating additional future weather information (State2) on the model's learning efficiency and decision-making quality.

The hyper-parameters of the DQN model are presented in Table 4. For training the Q-Network, we utilised a time step of 20 min. Additionally, we used data from January 2023 as meteorological data to train the agent.

The training process is illustrated in Fig. 5. As indicated by the graph, after 500,000 training steps, the rewards for all the cases tend to converge. The results indicate that the rewards for RL3 is significantly higher than the others. This is because RL1&2 only utilised State1, which is related to the current status and cannot effectively predict upcoming weather changes to proactively control the air conditioning.

Subsequently, we examine the performance of the trained models for each case using the test data. We tested the aforementioned two cases using data from two days in January 2024: cloudy (day 1) and sunny (day 2) days.

Table 5
Summary of comparisons in the test environment.

	Mode	Comfort time ratio	EC [kWh]
Baseline1	setpoint = 23	84%	4.06
Baseline2	setpoint = 22	58%	3.89
RL1	RL trained with rated power	83%	3.70
RL2	RL trained with COP model	84%	3.39
RL3	RL trained with COP model and weather forecast data	86%	2.77

In the test environment, the results of the comparison are summarized in Table 5, while the detailed comparison is illustrated in Fig. 6. We observed that in Baseline1 (setpoint = 23 °C), the room temperature remained relatively stable at 23 °C. However, since the heating load was below the air conditioner's minimum capacity at this time, the air conditioner was continuously in an intermittent operating state. This led to a high total energy consumption (EC) of 4.06 kWh over two days. In Baseline2 (setpoint = 22 °C), when we lowered the set temperature by 1 °C, the time the room temperature was below 22 °C increased due to the air conditioner's own PID control. This resulted in a 26% decrease in the comfort time ratio, and the energy-saving effect was limited, as indicated by the EC results. On the other hand, in the cases based on DRL, the agent adopted a threshold-based operation strategy, enabled the room temperature to fluctuate within the comfort zone ranging from 22 to 26 °C. This approach ensures that the air conditioner predominantly operates within the high-COP compliance range, thereby reducing the frequency of on-off cycles of the compressor and enhancing efficiency. Subsequently, we examined the test performance of each case.

First, based on the results from the previous study's case RL1, a comparison reveals that RL1 achieves the same high level of comfort time ratio as Baseline1 while reducing energy consumption to 3.70 kWh. This performance surpasses the aforementioned two baselines. However, as the air conditioner can only operate at its rated capacity, it is not possible to maintain the unit in a high COP state at all times. To address this issue, we proceed to analyze RL2 and RL3, which utilize a detailed COP model.

Due to the adoption of a detailed COP model in RL2 and RL3, the agent tends to operate the air conditioning during the mid-load, high-performance phases, which significantly reduces energy consumption. Furthermore, through a comparison between RL2 and RL3, we observed that after integrating predictive state, State2, into the model, the agent utilizes this predictive data to make informed decisions. This effectively leads to the proactive shutdown of the air conditioning on sunny days to prevent overheating. Additionally, following the assumed waking time of residents at 9 AM, the indoor temperature is restored to a comfortable range.

From the aforementioned comparison, we conclude that the adoption of a detailed COP model along with the introduction of the predictive state (State2), can effectively enhance the comfort of high-insulation residences, while reducing energy consumption by approximately 32%.

3.4. Back-end network training

To realize the planned heat input determined by the front-end network at the user-end, we learned the relationship between the air conditioner's heat input and the set temperature under various indoor and outdoor conditions through the back-end network. This learning process is a classification task within supervised learning. The features include the global horizontal solar irradiation, input heat as well as the indoor and outdoor temperatures.

In general, acquiring training data for air-conditioning systems requires extensive warm-up operations, often spanning several months. Such a procedure is resource-intensive and time-consuming. In our approach, we employed GANs for data augmentation, significantly curtailing the duration required for warm-up operations.

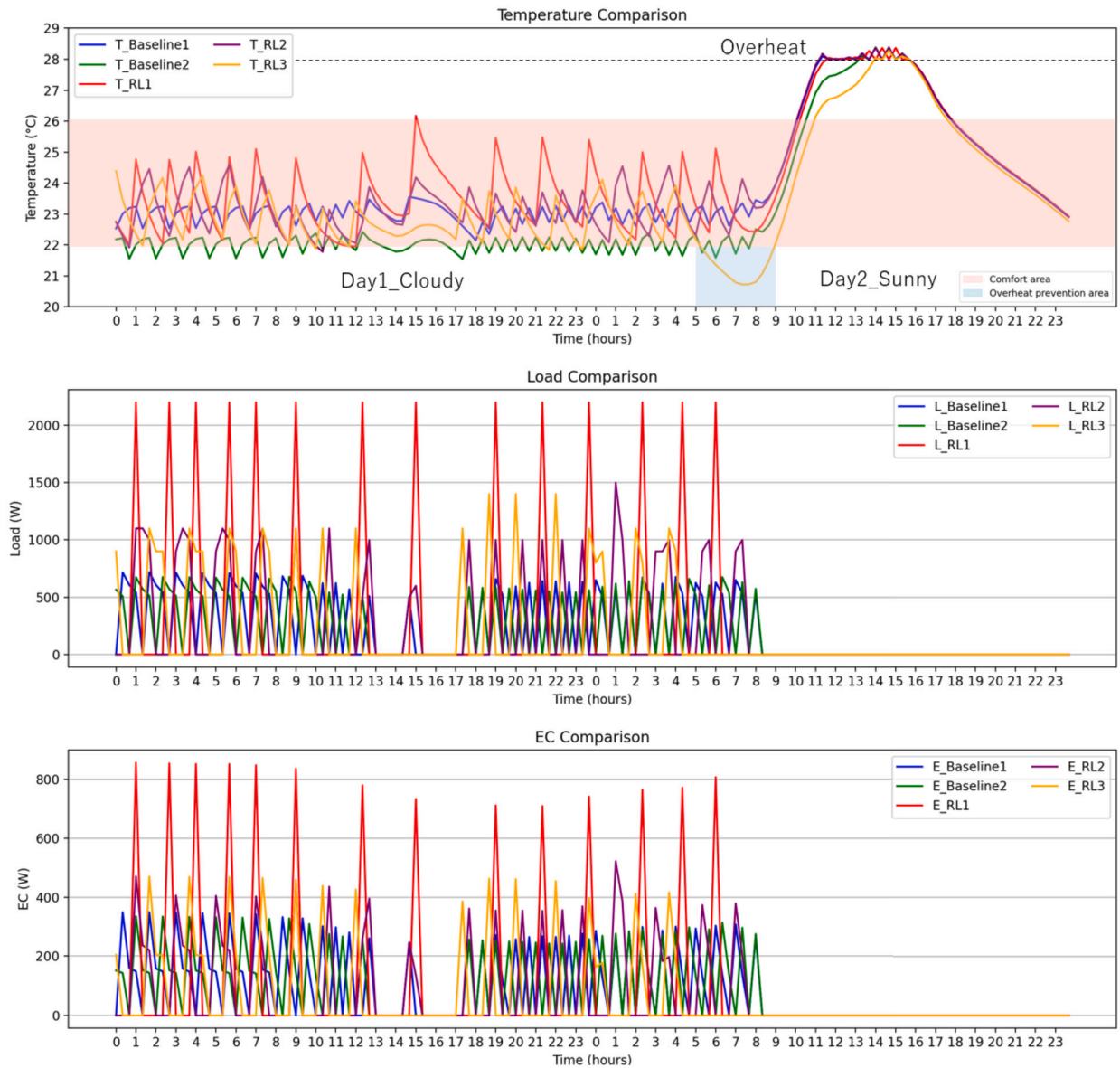


Fig. 6. Control performance in test environment.

A six-day empirical testing phase was conducted in our rooftop laboratory. On an hourly basis, we randomly selected set temperatures ranging from 22 to 25 °C. This strategy enabled us to efficiently amass an array of data points across a spectrum of operational conditions.

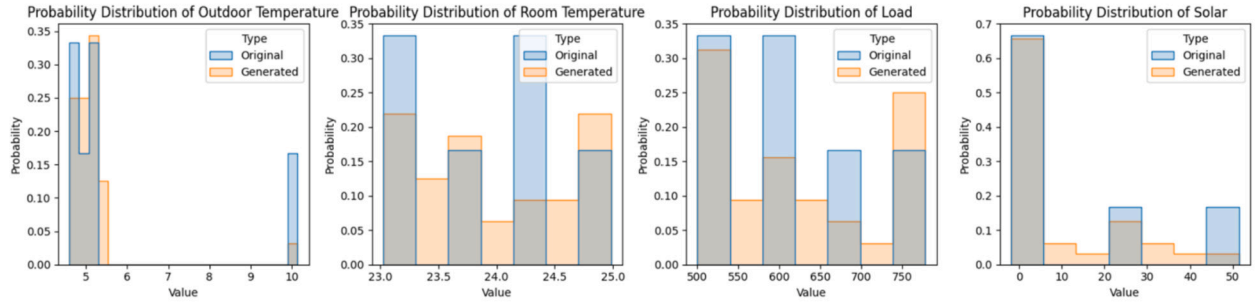
We adopted the same 20-in interval as used in the front-end network for data collection, averaging the heat input of the air conditioner. We can achieve zero heat input by turning off the air conditioning compressor, which can be realized either by switching off the air conditioner or by lowering its set temperature. However, in the case of this particular air conditioner, even when the set temperature is lowered and the compressor stops running, the fan remains operational, leading to increased electricity consumption. Therefore, in our approach, we opted to turn off the air conditioning to achieve zero heat input. We also implemented a filter to exclude data collected during instances of overheating of the room or when the set temperature was lower than the current temperature, leading to the compressor being turned off.

Our dataset was collected at 20-minute intervals, totaling 432 data points. However, due to the aforementioned reasons (zero heat input), only 83 data points remained after filtering. We divided the dataset into training and testing sets in a 70:30 ratio. The filtered data ex-

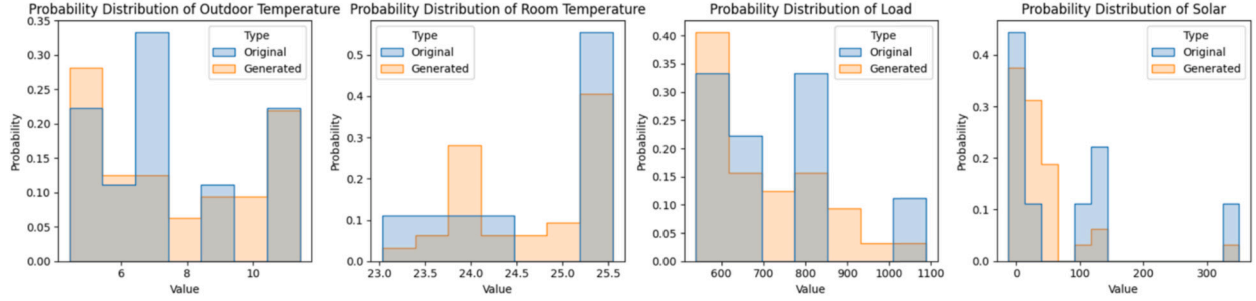
hibited extreme label imbalance, predominantly consisting of high set temperature outcomes (25 °C). This severe imbalance is not conducive to effective training in supervised learning. Consequently, we employed Generative Adversarial Networks (GANs) to generate synthetic data for set temperatures of 22, 23, and 24 °C, thus achieving a balanced distribution of labels in the new synthetic dataset. Fig. 7 illustrates the distribution of attributes in the synthetic dataset compared to the original dataset. Ultimately, we used this synthetic dataset for training purposes.

The comparison of results is presented in Table 6. To highlight the superiority of GANs, traditional data augmentation methods such as Synthetic Minority Over sampling Technique (SMOTE) and Adaptive Synthetic sampling (ADASYN) were also included in the analysis. It was observed that using SMOTE and ADASYN did not significantly improve the accuracy in the test set due to the limited sample size, which resulted in insufficient diversity in the data generated by traditional interpolation methods and increased the risk of overfitting to some extent. However, with the implementation of GANs for data augmentation, the accuracy was enhanced to 80%. Additionally, the recall rates for all categories showed improvements. Therefore, we conclude that using GANs

• Setpoint=22°C



• Setpoint=23°C



• Setpoint=24°C

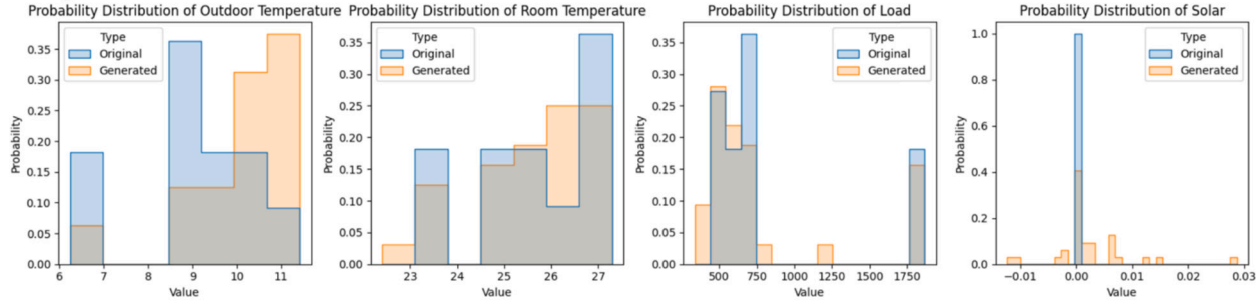


Fig. 7. Comparison between the distribution of attributes.

Table 6

Classification results from testing on the test dataset.

Database	Recall (22 °C)	Recall (23 °C)	Recall (24 °C)	Recall (25 °C)	Average Accuracy
Original data	1.00	0.30	0.75	1.00	0.68
SMOTE	1.00	0.30	0.75	1.00	0.68
ADASYN	1.00	0.30	1.00	0.80	0.68
GANs	1.00	0.50	1.00	1.00	0.80

for data augmentation increases the efficiency of data utilization and provides robust accuracy.

4. Implementation and verification

In this section, we detail the implementation of the control network, as reviewed in the previous chapter, onto a Raspberry Pi platform. This network was then deployed in the target building for a practical evaluation. Subsequently, we conducted a comparative analysis against a traditional baseline, wherein the set temperature remained constant. This comparative study was aimed at highlighting the enhancements in efficiency and performance of our control network when applied to a real-world setting.

4.1. Control process of the rooftop laboratory

The control process is delineated in Fig. 8. Initially, the trained control model was implemented on a Raspberry Pi platform, utilizing real-

time measured data (state1) along with weather forecast data (state2) as inputs to the front-end network to determine the heat input for the air-conditioning system. Subsequently, based on state1 and the planned heat input, the back-end network ascertains the specific control actions for the air-conditioning system (either setpoint or off). Finally, control over the air conditioning is achieved through the Echonet Lite protocol, enabling real-time adjustments based on the network's output.

4.2. Empirical results and analysis

We conducted a practical performance test of our control model in January. The weather conditions during this testing phase are presented in Table 7. To evaluate the effectiveness of the control model under similar weather conditions, we selected three distinct cases with varying levels of solar radiation, namely, medium, high, and low, and compared their performances against a baseline scenario where the set temperature remained unchanged.

The energy consumption comparison results are presented in Table 8, and the detailed comparisons among individual cases are as follows:

- (1) The detailed comparison results for Case A_Medium solar are shown in Fig. 9. First, we introduce the baseline. From the results of room temperature distribution, we observed that when the set temperature remains constant, the room temperature fluctuates between 24 to 25 °C. At approximately 10 A.M., owing to the influence of solar radiation, the room temperature exceeds the set temperature, causing the compressor of the air conditioner to stop running. From

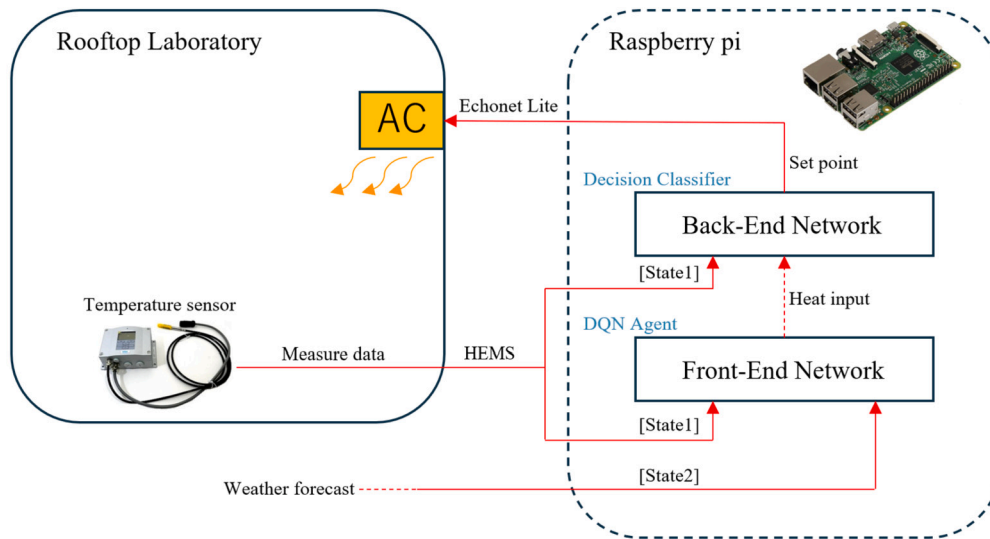


Fig. 8. Control flow in target building.

Table 7
Meteorological conditions.

	Date	Mode	Average ambient temperature [°C]	Total global horizontal solar irradiation [kWh/m ²]	Sunshine duration [h]
CaseB_Baseline	2024/1/6	set = 22	11	3.1	8.9
CaseA_Baseline	2024/1/7	set = 22	9.9	2.28	3.2
CaseC_Baseline	2024/1/21	set = 22	8.1	0.62	6.5
CaseA_DRL	2024/1/18	RL	9.8	2.25	4
CaseC_DRL	2024/1/20	RL	7	0.63	0
CaseB_DRL	2024/1/23	RL	10.9	3.24	7.9

Table 8
Result of the implementation.

	Date	Mode	AC heat input [kWh]	Overheat [kWh]	Comfort time ratio	EC [kWh]	Average COP
CaseA_Baseline	2024/1/7	set = 22	4.65	0.79	79.30%	2.21	2.10
CaseB_Baseline	2024/1/6	set = 22	3.58	4.17	58.40%	1.71	2.09
CaseC_Baseline	2024/1/21	set = 22	10.26	0	100.00%	3.82	2.69
CaseA_DRL	2024/1/18	DRL	3.67	0.17	74.65%	1.36	2.70
CaseB_DRL	2024/1/23	DRL	2.73	2.21	63.06%	1	2.73
CaseC_DRL	2024/1/20	DRL	8.81	0	86.10%	3.24	2.72

approximately 11 A.M. to 2 P.M., the indoor environment experiences overheating. The compressor of the air-conditioning system restarts operation at approximately 5 P.M. The percentage of time within the comfort zone of room temperature over the course of the day was 79.3%. Subsequently, when employing the DRL control, we observed that the model could effectively detect overheating conditions and pre-emptively shut down the air-conditioning system, reducing the heat input by approximately 1 kWh and decreasing overheating by 0.62 kWh. Owing to the substantial heat input during air-conditioner startup and the slightly smaller area of the rooftop laboratory, the temperature slightly exceeded the upper limit of the comfort zone, reaching 74.7%. Considering the daily average COP of the air conditioner, we observed that with the adoption of the DRL control, the average COP increased from 2.1 to 2.7. Considering the reduction in heat input and aforementioned improvement in COP, the electricity consumption of the air conditioner decreased by 38%, reaching 1.36 kWh.

(2) The detailed comparison results for Case B_High solar are shown in Fig. 10. A detailed comparison revealed that the conclusions drawn

from Case B were roughly similar to those from Case A. By utilising a control model that can anticipate overheating in advance, the heat input of the air-conditioning system reduced by 0.51 kWh. This approach also significantly decreased the overheating by more than half, namely, to 2.22 kWh. Furthermore, the intermittent operation of the air-conditioning system increased the COP from 2.1 to 2.9. Owing to these factors, the daily energy consumption reduced from 1.71 to 1.05 kWh, achieving energy savings of 39%.

(3) The detailed comparison results for Case C_Low solar are shown in Fig. 11. Owing to the rainy weather on both days of Case C, the solar radiation was minimal, resulting in the continuous operation of the air-conditioning system without any overheating incidents. The energy consumption comparison between the two scenarios indicated a marginal difference in the COP. This is because the average heat input in the baseline scenario (428W) was higher than that under the DRL control (327W). Considering that the COP of the air conditioner is significantly influenced by the load rate in low-load conditions, the baseline COP was adjusted to match the compliance rate of the DRL scenario, resulting in a corrected COP of 2.38. The

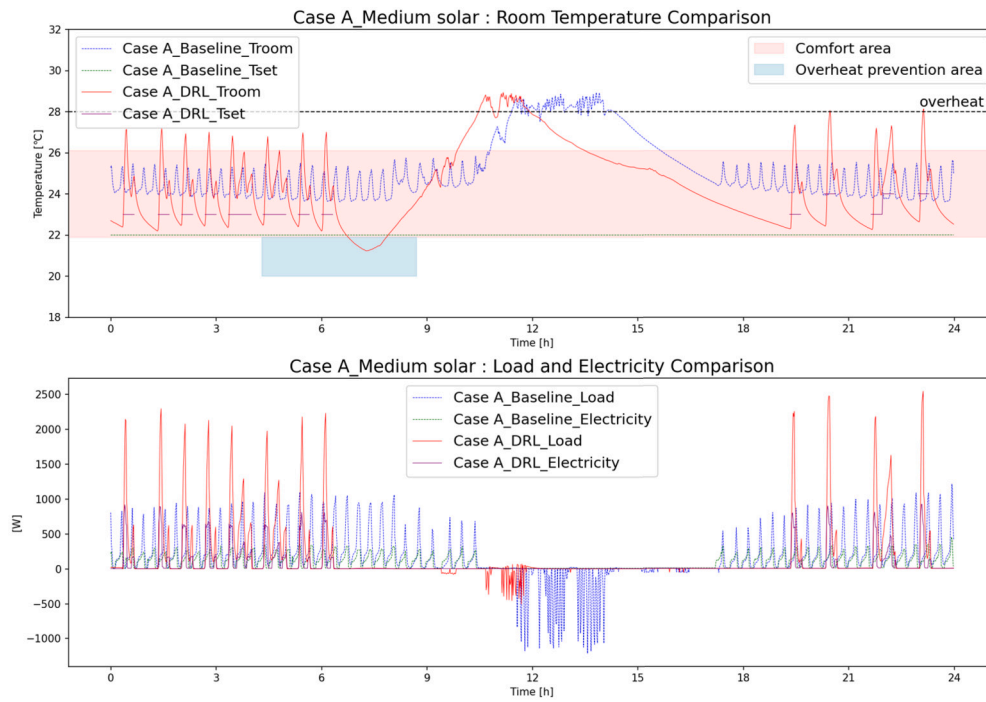


Fig. 9. Case A_Medium solar.

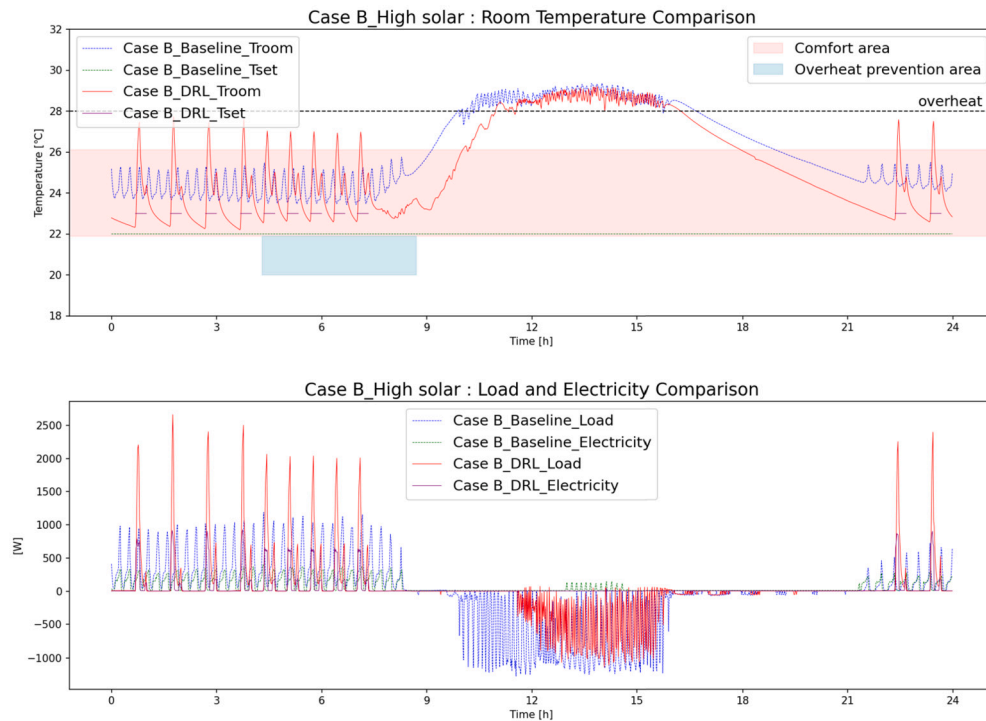


Fig. 10. Case B_High solar.

comparison revealed that adopting the DRL control enhanced the efficiency of the air conditioner by 14.3%.

5. Conclusion and future work

In this study, we introduced a DRL-based real-time & prediction full-stack control model to enhance energy efficiency and reduce consumption of air conditioning systems in high-insulation residential buildings. During the training process, we utilized a detailed air conditioning COP

model and achieved precise control of the air conditioning by integrating weather forecast data with DRL. The results indicated that under low-load conditions, by employing a threshold-like operating method, we can reduce the number of start-stop cycles of the compressor, thus improving the COP of the air conditioner in low-load areas. Additionally, by pre-emptively turning off the air conditioning, overheating could be mitigated to achieve energy savings. Ultimately, experimental evidence demonstrated that utilising this control model leads to a reduction in energy consumption by approximately 40%. In terms of the applica-

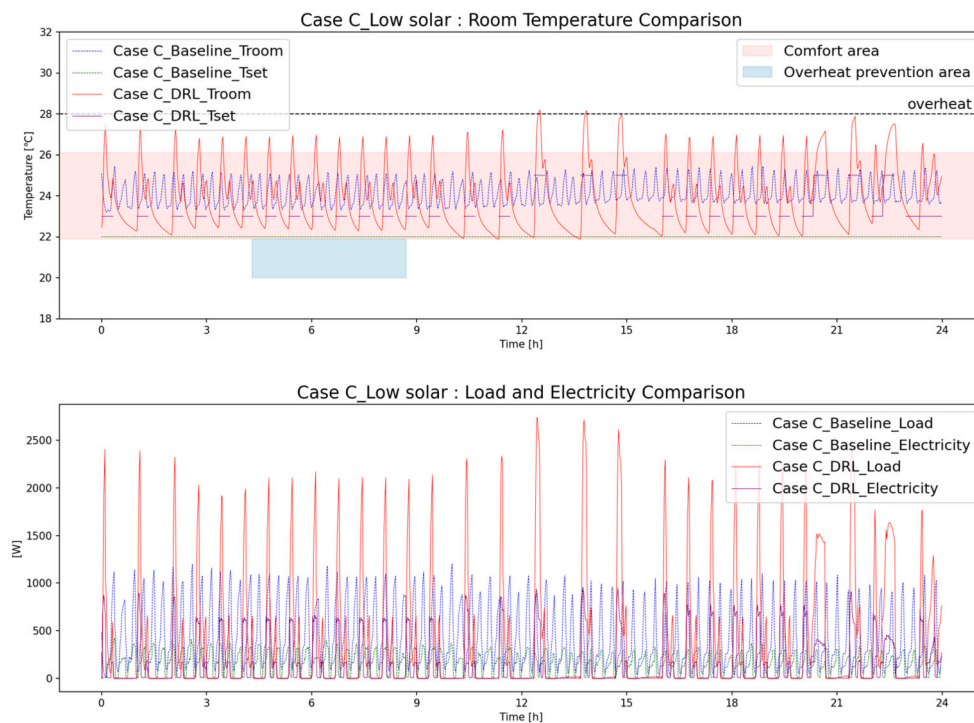


Fig. 11. Case C_Low solar.

tion, we employed a back-end network based on supervised learning and data augmentation using GANs, effectively utilising warm-up operational data to minimise the warm-up duration.

For future research directions, we anticipate that augmenting empirical data collected post-deployment and continuously improving the back-end network will enable persistent learning and adaptation. Additionally, our future objectives include enhancing the model's scalability and generalizability, focusing on its application across a broad spectrum of architectural styles and climatic conditions. Furthermore, we aim to deploy this model in actual residential settings to rigorously evaluate its performance and efficacy in real-world scenarios.

CRedit authorship contribution statement

Luning Sun: Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zehuan Hu:** Methodology, Data curation. **Masayuki Mae:** Supervision. **Taiji Imaizumi:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported by JSPS KAKENHI Grant Numbers JP23K26258, JP23H01564.

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